From Assisted Input to Enforced Autonomy: A Conceptual Design Framework for Support Withdrawal in L2 Reading

Abstract

Digital environments for second language (L2) reading increasingly prioritize immediate comprehension through in-flow lexical support such as glosses, pop-up dictionaries, and on-demand translation. While these tools effectively reduce processing difficulty and sustain reading flow, they also render assistance continuously available, making it methodologically difficult to distinguish autonomous processing from efficient support use. Although learner autonomy is frequently invoked as a long-term goal of assisted reading, the transition from supported to unsupported processing is rarely specified, enforced, or isolated as a design variable.

This article introduces Supported Two-Phase Reading (STR) as a design framework that formalizes the withdrawal of support as an explicit, mandatory transition within the reading process. STR specifies a fixed sequence in which learners first read a text with full in-flow support and then immediately reread the identical text with all support categorically removed. By holding text, content, and prior resolution constant, the framework renders the withdrawal event itself analytically observable, rather than treating supported and unsupported reading as alternative instructional conditions.

The contribution of this article is conceptual and methodological rather than empirical. STR reframes autonomy not as a learner choice or assumed outcome, but as a design-imposed processing condition that can be examined at the moment of transition. The framework introduces protocol-bound, transition-based descriptors intended to capture learners' adjustment when support is withdrawn, while explicitly limiting their interpretive scope. No empirical data are reported, and no claims are made regarding learning outcomes. The article offers a formal design specification intended to support future empirical investigation of enforced support withdrawal in CALL research.

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1. Introduction

Digital tools for L2 reading have undergone rapid development over the past decade. Contemporary platforms routinely integrate lexical glosses, pop-up dictionaries, morphological hints, and automatic translation directly into the reading interface. These features substantially reduce the cognitive burden associated with unknown vocabulary and grammatical forms, allowing learners to maintain comprehension and reading flow even when lexical coverage is limited. From a design perspective, such systems have been highly successful in optimizing accessibility and engagement, enabling learners to work with authentic or semi-authentic texts earlier than would otherwise be feasible.

At the same time, the proliferation of support-rich reading environments has reinforced a characteristic pattern in CALL research: design decisions concerning support availability and sequencing are often treated as self-evident implementation choices rather than as objects of explicit methodological analysis. Interface configurations are commonly embedded directly into experimental conditions, with limited discussion of why a particular support architecture is analytically necessary as opposed to merely convenient. As a result, critical design dimensions tend to be naturalized, and the conceptual space between pedagogical intention and empirical observability remains under-articulated.

One such under-specified dimension concerns the withdrawal of reading support. While learner autonomy is widely referenced as a desired outcome of assisted reading, the conditions under which autonomy is expected to occur are rarely instantiated at the level of design. In most digital reading environments, support remains continuously available and learner-controlled, leaving the transition from supported comprehension to autonomous processing implicit and optional. Consequently, autonomy is typically inferred post hoc from successful task performance rather than treated as an observable processing condition within the reading event itself.

This creates a methodological blind spot. As long as support remains continuously accessible, successful reading performance cannot be analytically distinguished from efficient use of external aids. The absence of overt lookup behavior does not constitute evidence of autonomous processing, nor does fluent comprehension guarantee reliance on internal representations rather than interface-mediated support. Under such conditions, autonomy functions as a presumed endpoint rather than as a state that can be directly examined.

This article argues that the absence of a protocol in which support withdrawal is imposed by design—rather than left to learner discretion—constitutes a conceptual gap in L2 reading research. While CALL studies have extensively examined different types of glosses, dictionaries, and support modalities, comparatively little attention has been paid to the withdrawal event itself: the moment at which assistance is removed and learners are required to process text without external support. Support-free reading is typically treated either as a separate experimental condition or as an assumed outcome of prolonged exposure, rather than as a designed transition that can be isolated and analyzed.

From a design perspective, this omission has non-trivial consequences. When support remains permanently available, the distinction between assisted and autonomous processing is methodologically obscured. Conversely, when support is removed without prior stabilization of comprehension, processing breakdown or disengagement may occur. What is missing is not another comparison of supported versus unsupported reading, but a principled way to formalize the withdrawal event itself as a design primitive—one that renders the transition from supported to unsupported processing explicit, mandatory, and analytically observable.

The present article addresses this gap by proposing Supported Two-Phase Reading (STR) as a constrained design protocol that formalizes enforced support withdrawal as an architectural condition. STR is not introduced as a teaching method, nor as a claim about instructional effectiveness. Instead, it is presented as a minimal design framework whose primary contribution lies in making autonomy observable as a protocol-defined processing condition rather than treating it as an inferred outcome.

STR specifies a fixed two-phase sequence applied to the same text. In Phase I, learners read with full in-flow support to ensure stable comprehension. In Phase II, they immediately reread the identical text with all support categorically removed. The transition between phases is neither optional nor learner-controlled; it is defined as a design-imposed constraint. By enforcing this transition, STR renders the withdrawal of support analytically accessible rather than leaving it as an implicit background assumption.

This framing shifts analytic attention away from the mere presence or absence of support and toward learners' adjustment at the point of withdrawal. Because text, content, and prior resolution are held constant across phases, observed differences in processing time, task-level performance, or subjective effort can be interpreted as responses to the withdrawal event itself rather than to changes in textual difficulty. In this sense, STR treats the transition—not the individual phases—as the primary unit of analysis.

STR is deliberately positioned as a minimal, design-driven construct. It differs from traditional repeated reading, which typically involves multiple unguided encounters aimed at fluency development, and from learner-controlled support systems, where autonomy remains a matter of choice. In STR, autonomy is structurally enforced. While the framework is compatible with accounts that emphasize retrieval-related processing demands, it does not claim to implement delayed or optimally spaced retrieval practice. Instead, it offers a parsimonious way to instantiate retrieval-like demands within the continuous flow of reading under conditions of maximal design control.

The contribution of this article is therefore conceptual rather than empirical. It does not claim that STR produces superior learning outcomes, nor does it attempt to validate psychological constructs. Rather, it provides a formalized design object for researchers interested in examining enforced support withdrawal as a design variable in CALL environments, while explicitly limiting the interpretive scope of any resulting measures to the confines of the protocol.

2. Background and Design Gap

2.1 Assisted Reading in CALL

Research on assisted reading in CALL has consistently demonstrated that lexical and grammatical support can substantially facilitate comprehension during L2 reading. Common implementations include static glosses, pop-up dictionaries, marginal annotations, and increasingly, adaptive systems that tailor support to learner profiles or inferred proficiency. Across these formats, the primary design objective is to reduce processing difficulty by making meaning rapidly accessible at the point of need.

Glossing, in particular, has been extensively studied as a means of supporting vocabulary acquisition and text comprehension. Whether presented as L1 translations, L2 paraphrases, or multimedia annotations, glosses allow learners to resolve lexical uncertainty without abandoning the reading task. Similarly, pop-up dictionaries embedded in digital texts enable on-demand lookup while preserving linear reading order more effectively than external reference tools. Adaptive support systems further refine this approach by attempting to predict which items require assistance and adjusting support density accordingly.

A shared characteristic of these designs is that **support availability is typically continuous and learner-controlled**. Learners decide whether and when to consult a gloss, click on a word, or request additional information. From a usability perspective, this design choice aligns with principles of learner autonomy and individualized pacing. From a research perspective, however, it introduces a degree of indeterminacy: the extent to which learners rely on support, avoid it, or strategically alternate between recognition and retrieval is rarely constrained by the system itself.

As a result, assisted reading environments in CALL tend to prioritize **accessibility and flow**, ensuring that comprehension breakdowns are quickly repaired. What they do not systematically specify is how, or when, learners are expected to transition away from reliance on external support.

2.2 The Frictionless Reading Paradigm

The prevalence of always-available support has given rise to what can be described as a *frictionless reading paradigm* in digital L2 environments. In this paradigm, reading is optimized to minimize interruption, effort, and uncertainty. Unknown items are resolved instantly, often with a single click, and the system is designed to ensure that comprehension is rarely jeopardized.

While this design is highly effective for maintaining engagement, it also introduces methodological challenges. Lookup behavior becomes a persistent confound: when support is always accessible, it is difficult to determine whether comprehension reflects internalized knowledge or moment-to-moment reliance on external aids. Successful reading under such conditions does not necessarily indicate autonomous processing, but may instead reflect efficient navigation of the interface.

Moreover, autonomy in these systems is typically **assumed rather than enforced**. Learners are expected to gradually reduce their dependence on support as proficiency increases, but this reduction is neither required nor structurally encouraged. In practice, many learners continue to rely on glosses and pop-up dictionaries even for familiar items, particularly when the cost of lookup is negligible. Consequently, the distinction between supported and autonomous reading remains blurred.

From a design standpoint, frictionless reading environments treat autonomy as an emergent outcome of prolonged exposure rather than as a condition that can be explicitly instantiated. This assumption limits the extent to which CALL research can examine the cognitive and behavioral consequences of reading without support.

2.3 The Unspecified Transition Problem

Despite widespread acknowledgement that learners should eventually read without assistance, the transition from supported to unsupported reading is rarely specified as a design element. In much of the literature, support withdrawal remains implicit: glosses are either available throughout a task or absent altogether, while the act of removing support is not itself modeled as part of the instructional or experimental architecture.

When rereading occurs, it is typically optional and learner-initiated. Learners may choose to reread a text without support, but they are equally free to continue consulting glosses or to disengage from the task. Under such conditions, rereading does not constitute a controlled design event but rather an uncontrolled strategy choice, confounding autonomous processing with individual differences in effort, persistence, or interface use.

This design pattern creates a conceptual and methodological gap. Although autonomy is frequently discussed as a desirable learning outcome, the conditions under which autonomous processing is required are rarely enforced at the level of system architecture. Without a design that mandates a transition to unsupported processing, it remains difficult to observe how learners adjust when assistance is no longer available, or to distinguish recognition-based comprehension supported by external cues from processing that relies on internal representations.

As a result, CALL research has extensively modeled supported processing, but has not systematically modeled the transition from supported to autonomous processing within a single reading sequence. The absence of a design-defined transition constrains both theoretical interpretation and empirical analysis of how autonomy manifests in digital reading environments.

Thus, the critical gap is not the study of autonomy as an outcome, nor the study of support as a tool. The gap lies in the systematic omission of the withdrawal event itself as a manipulable design primitive. STR addresses this omission by reifying support withdrawal as a mandatory, binary transition, thereby transforming an implicit assumption into an explicit, analytically accessible design variable.

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3. Conceptual Positioning of STR

The purpose of this section is to position Supported Two-Phase Reading (STR) as a design framework and delimit its intended interpretive scope. STR is not introduced as a new instructional method competing with existing approaches, but as a design framework intended to make support withdrawal explicit and analyzable.

3.1 What STR Is Not

Before specifying the components of the STR framework, it is necessary to clarify what the present proposal deliberately does not attempt to do. These exclusions are not limitations in the conventional sense, but defining properties of the design stance adopted in this article.

First, STR is not a teaching method. It does not prescribe curricular objectives, instructional sequencing beyond the two-phase constraint, or pedagogical content selection. The framework does not aim to optimize learning outcomes, nor does it offer guidance regarding frequency of use, learner grouping, or long-term instructional integration. STR specifies a local, protocol-level design intervention rather than a comprehensive pedagogical approach.

Second, STR is not a material or interface format. Although it can be instantiated in digital or non-digital environments, the framework is not tied to a particular presentation style, glossing technique, or user interface. Its defining characteristic lies in the enforced temporal organization of reading phases, rather than in the visual, linguistic, or technological realization of support.

Third, STR is not a claim of instructional superiority. The framework does not argue that two-phase reading is more effective than extensive reading, traditional glossing, or learner-controlled support systems. Any claims regarding relative effectiveness, retention, or proficiency development would require empirical testing under conditions that extend beyond the scope of the present article.

Crucially, STR is intentionally underpowered with respect to outcome claims. The framework is designed to formalize a transition mechanism rather than to maximize or demonstrate learning gains. Its purpose is not to estimate effects, but to render the consequences of enforced support withdrawal observable within a tightly constrained design space. As such, STR prioritizes analytic interpretability of the transition over sensitivity to distal or longitudinal learning outcomes.

Taken together, these constraints position STR as a design object rather than an intervention. The framework is proposed to support principled investigation of support withdrawal as a design variable, not to function as a ready-made instructional solution or an empirically validated treatment.

3.2 What STR Is

STR represents a conceptual reorientation in how autonomy is approached in L2 reading research. Rather than treating autonomy as a learner trait, disposition, or voluntary strategy choice, STR treats autonomy as a design-imposed processing condition. This reorientation shifts the research focus from asking whether learners become autonomous to examining what occurs, observably and measurably, when autonomous processing is architecturally required.

In this sense, STR moves the analytic question from *Do learners choose to read autonomously?* to *What happens when autonomy is the only available mode of processing?* The framework does not assume that such enforced autonomy is optimal or beneficial; it assumes only that it must be instantiated by design in order to be studied as a distinct processing condition.

Table 1 summarizes the design scope and explicit boundary conditions of the STR framework.

Design dimension	Scope addressed by STR	Deliberately out of scope
Text length	600–800-word narrative	< 400 or > 1 500 words
Learner age	Adult L2 users (18+)	School-age learners
Interval between phases	Immediate rereading (0 min)	Delayed > 1 day
Skill domain	Reading comprehension	Writing, listening, speaking
Support fading schedule	Binary (100% → 0%)	Gradual or adaptive fading

STR is proposed as a design protocol consisting of a minimal, fixed sequence of actions applied to the same text. The protocol requires learners to engage in two consecutive encounters: an initial supported reading followed immediately by an unsupported rereading of the identical text.

The defining characteristic of STR is that the transition between phases is mandatory and categorical. Support is fully available in Phase I and fully absent in Phase II. Learners cannot partially access support, defer the transition, or reverse it once Phase II has begun. In this sense, autonomy is not a learner decision but a design-imposed processing condition.

By constraining learner options, STR transforms support withdrawal from an implicit assumption into an explicit design-defined event. This transformation renders the transition analytically accessible at the level of protocol specification and within-sequence comparison, allowing researchers to examine learners' adjustment to enforced autonomy without conflating autonomous processing with individual strategy choice or interface behavior.

3.3 Relation to Existing Paradigms

STR intersects with several established paradigms without fully belonging to any single one.

It is related to **assisted reading**, insofar as Phase I provides in-flow support to ensure stable comprehension. However, unlike typical assisted reading designs, support is temporally limited rather than continuously available.

It resembles **repeated reading** in that learners encounter the same text more than once. Unlike traditional repeated reading, however, repetition in STR is immediate, obligatory, and asymmetrical with respect to support availability.

STR also draws on the logic of **scaffolding**, particularly the idea that external assistance should be temporary. Yet the framework operationalizes scaffolding in a binary, non-adaptive manner suitable for constrained experimental and design-based research contexts.

Finally, STR is compatible with **retrieval-based learning accounts** at the level of mechanism. The unsupported rereading phase imposes retrieval-like processing demands by removing external cues. Importantly, STR does not claim to implement optimal retrieval practice, nor does it assume delayed consolidation. Retrieval is treated as a plausible processing mechanism, not as an outcome claim.

4. STR Framework Architecture

Supported Two-Phase Reading (STR) is defined by a minimal architectural sequence rather than by instructional content or material design. The framework specifies *when* and *how* support is available, and more importantly, *when it is not*. This section describes the structure of the protocol and clarifies the rationale behind its key design decisions.

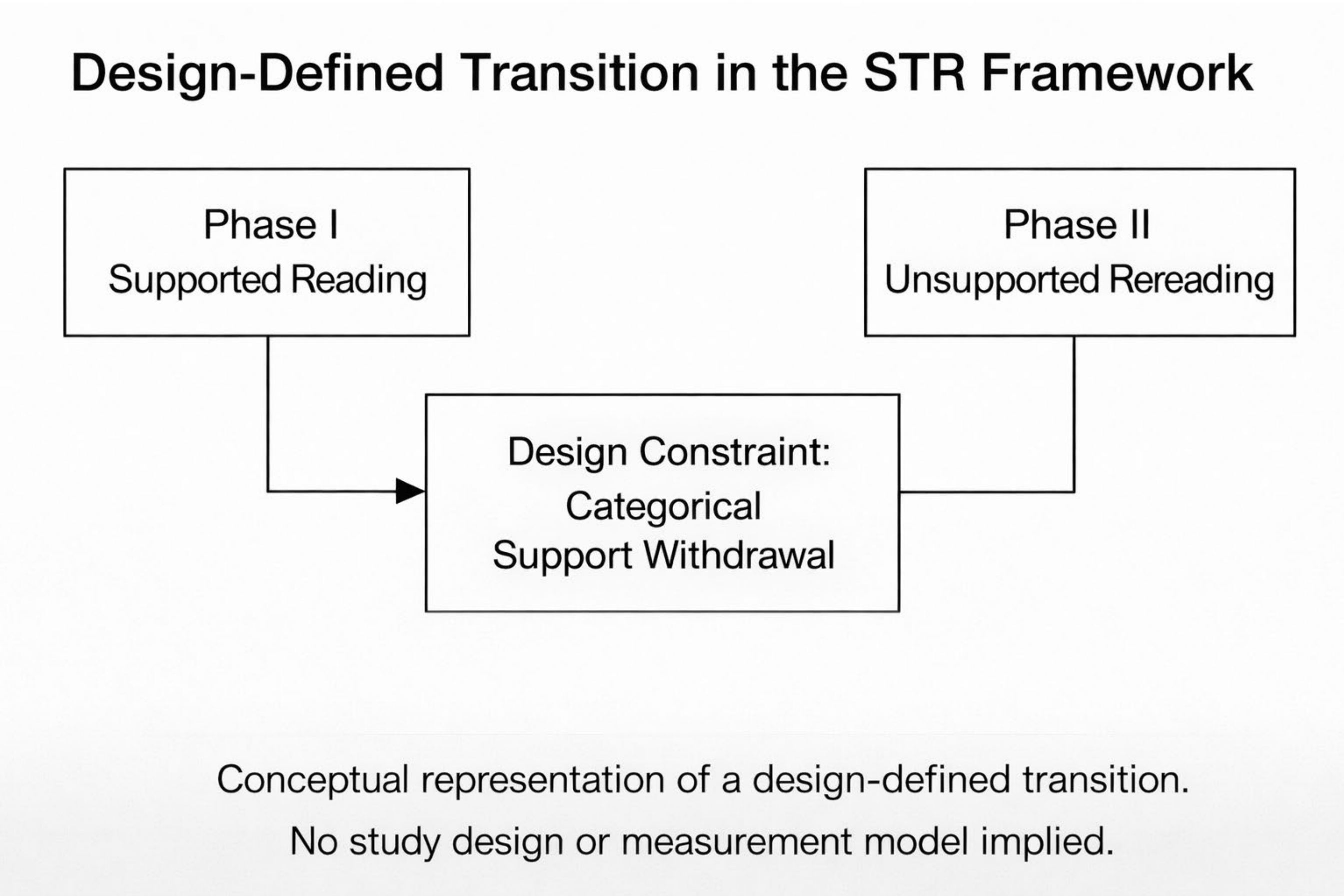

Figure 1. Conceptual design constraint highlighting the withdrawal event in the STR protocol. The figure depicts a schematic reading sequence centered on the categorical withdrawal of support between two consecutive phases. It is intended solely as a conceptual representation of a design constraint and does not imply an experimental design, study flow, measurement model, or data-collection scheme.

4.1 Phase I: Supported Reading

Phase I consists of a single encounter with the target text under conditions of full in-flow support. Lexical and grammatical assistance is integrated directly into the text stream, allowing learners to access meaning without interrupting linear reading. Crucially, Phase I does not permit lookup actions, navigation to external resources, or interaction with the support beyond passive availability.

The design goal of Phase I is not memorization or testing, but **stable comprehension and initial encoding**. By minimizing extraneous processing demands associated with search and navigation, Phase I aims to support the construction of a coherent situation model and the initial mapping between form and meaning. Learners may attend selectively to the support or attempt to process the text autonomously, but the availability of assistance ensures that comprehension breakdowns do not derail processing.

Importantly, Phase I does not require learners to engage in retrieval. Recognition-based processing is sufficient at this stage, and no assumption is made that learners will retain specific lexical items after the first encounter. Phase I is thus framed as a preparatory phase that establishes a common baseline of understanding prior to the withdrawal of support.

4.2 Binary Fading as a Design Choice

A defining feature of STR is the **binary removal of support** between phases. Rather than gradually reducing assistance or adapting support density based on learner performance, STR implements a categorical shift from full support (100%) to no support (0%).

This decision is motivated by **experimental tractability rather than pedagogical optimality**. Gradual or adaptive fading introduces multiple interacting variables—such as timing, granularity, and learner sensitivity—that complicate interpretation. By contrast, binary fading allows support availability to be treated as a clearly manipulable design variable, facilitating analysis of its immediate consequences.

The framework does not claim that binary fading is instructionally ideal, nor that it mirrors expert scaffolding in classroom contexts. Instead, it represents a deliberate simplification that enables researchers to isolate the effects of support withdrawal within a constrained design. STR should therefore be understood as a research-oriented protocol, not as a prescription for instructional sequencing.

4.3 Phase II: Unsupported Rereading

Phase II requires learners to reread the **identical text** encountered in Phase I, but with all external support removed. No glosses, translations, or hints are available, and learners are not permitted to return to Phase I once Phase II has begun.

Because the text remains unchanged, Phase II does not introduce new linguistic material or additional content complexity. The sole difference between phases is the availability of support. Under these conditions, learners must rely on their internal representations to reconstruct meaning, engaging in autonomous processing to the extent that prior encoding permits.

The unsupported rereading phase thus imposes **forced autonomous processing**. Unlike learner-controlled environments, where autonomy is optional, STR enforces autonomy by design. Learners cannot circumvent the absence of support through lookup or interface interaction, making their adjustment to unsupported processing observable.

4.4 Why the Transition Is the Unit of Analysis

In STR, the analytically relevant event is not supported reading or unsupported reading in isolation, but the **transition between them**. By holding the text constant and minimizing variation in learner characteristics within a session, the framework treats support availability as the primary manipulated variable.

This design allows researchers to examine how learners respond when assistance that was previously available is categorically withdrawn. Changes in comprehension, processing time, or subjective effort can be interpreted as consequences of the transition itself, rather than as artifacts of text difficulty or unfamiliar content.

By conceptualizing the transition as the unit of analysis, STR reframes support withdrawal from an implicit assumption into an explicit design-defined event. This shift enables more precise investigation of autonomy-related processing in L2 reading.

4.5 Illustrative Scenario: A Single-Text STR Cycle

The following scenario is a design illustration, not an example of empirical implementation. It is included solely to clarify how the Supported Two-Phase Reading (STR) framework can be instantiated while preserving its core design constraints.

This section provides an illustrative instantiation of STR using a single narrative text. The purpose of the scenario is not to demonstrate instructional effectiveness, nor to evaluate learning outcomes, but to make the internal structure of the protocol explicit. The scenario functions as a worked design illustration of the framework architecture described in Sections 4.1–4.4, showing how the transition from supported to unsupported reading is specified as a design-defined event.

Importantly, the scenario is not presented as pedagogical material and does not constitute empirical evidence. It serves to demonstrate how STR isolates the withdrawal of support as the focal design operation by holding text, task, and learner constant across phases.

No data, participants, or outcomes are implied by this scenario.

4.5.1 Context and Boundary Conditions

The illustrative scenario assumes the following conditions, which are intentionally constrained to avoid overgeneralization:

- Adult L2 learners engaged in self-directed reading
- Shared L1 background for the purpose of demonstrating L1-based glossing
- A single continuous narrative text of approximately 600–800 words
- Reading for general comprehension, without additional tasks or assessments

These boundary conditions reflect a common use case in digital reading environments while remaining narrow enough to preserve experimental tractability. The scenario does not assume classroom instruction, teacher mediation, or adaptive system behavior.

4.5.2 Phase I: Supported Reading (Illustrative Excerpt)

In Phase I, learners encounter the text with full, integrated linguistic support. Support is provided in-flow, directly following target lexical or grammatical items within the linear progression of the L2 text. No interaction is required to access support: learners do not click, hover, toggle, or search. All support is continuously visible.

The following excerpt is an illustrative example originating from industry practice and is provided solely to demonstrate the STR design constraints, specifically the implementation of in-flow support. It is adapted from a public-domain text and is non-evaluative; it does not constitute instructional material, empirical evidence, or a basis for outcome claims.

I have seldom heard (sotva jsem slyšel; *to hear, heard, heard – slyšet, zaslechnout, vnímat*) him call her differently (že by ji nazýval jinak; *to call – volat, nazývat, zvát; different – jiný, odlišný, rozdílný*). For him she is (pro něj je) a representation of her whole sex (ztělesněním celého ženského rodu; *representation – zastoupení, ztělesnění, reprezentace*). But don't get any ideas (ale nedělejte si žádné iluze; *to get an idea – dostat nápad, udělat si představu, přijít na myšlenku*), he wasn't in love with her (nebyl do ní zamilovaný; *to be in love with – být zamilovaný do, být uchvácen, být poblázněný*).

This format exemplifies several design constraints central to STR:

1. **In-flow support**: Linguistic assistance is spatially integrated with the L2 text, eliminating the need for external lookup or interface interaction.
2. **Visual primacy of L2**: The L2 text remains visually dominant, while support is visually differentiated (e.g., reduced contrast, color coding).
3. **Absence of learner control**: Learners cannot selectively hide, reveal, or request support. Support availability is determined entirely by the protocol.
4. **Comprehension-oriented processing**: Phase I is designed to stabilize comprehension and enable initial encoding by minimizing extraneous cognitive load associated with search and navigation.

Phase I does not impose a memorization requirement, nor does it include explicit prompts to attend to form. Its function within the STR framework is to ensure that comprehension is sufficiently supported so that the subsequent withdrawal of support constitutes a meaningful transition rather than a shift in task difficulty or text accessibility.

4.5.3 The Transition Event: Categorical Withdrawal of Support

Upon completion of Phase I, learners advance to Phase II through a mandatory transition. This transition is the central design-defined event in the STR framework.

Key characteristics of the transition include:

- **Categorical withdrawal**: All linguistic support is removed simultaneously (100% → 0%).
- **No backward navigation**: Learners cannot return to the supported version once the transition occurs.
- **Explicit phase boundary**: The system clearly signals that the same text will be re-presented without support.

Crucially, the transition does not introduce a new task, assessment, or instructional demand. The learner's objective remains unchanged: reading the text for comprehension. The only manipulated variable is the availability of external support.

By enforcing this transition architecturally rather than leaving it to learner choice, STR makes support withdrawal observable and analyzable. The framework thus treats withdrawal not as an assumed outcome of learning, but as a controlled design event.

4.5.4 Phase II: Unsupported Rereading

In Phase II, learners encounter the identical L2 text without any linguistic support. The text is word-for-word identical to that presented in Phase I, preserving lexical, syntactic, and discourse structure.

For example, the same excerpt appears as follows:

I have seldom heard him call her differently. For him she is a representation of her whole sex. But don't get any ideas, he wasn't in love with her.

No dictionaries, glosses, translations, or explanatory annotations are available. Interface affordances that would allow external lookup are either disabled or absent by design.

Phase II is not conceptualized as a test, nor does it function as a control condition. Instead, it constitutes a forced rereading under conditions of reduced external mediation. The learner is required to process the text autonomously, relying on internalized representations formed during Phase I.

4.5.5 Observable Outputs and Protocol-Level Measures

Although the present article does not report empirical data, the illustrative scenario clarifies what types of observable outputs the STR protocol makes available for analysis. These outputs are protocol-bound and do not correspond to stable psychological traits.

Examples include:

- **Reading time per phase**, allowing comparison of processing effort before and after support withdrawal
- **Self-rated comprehension**, collected immediately after each phase
- **Behavioral indicators of support dependence**, such as attempts to access L1 resources when blocked
- **Relative performance adjustment**, operationalized as within-text differences rather than between-group contrasts

These outputs become comparable across phases only because the STR design holds text, task, and learner constant while manipulating support availability.

4.5.6 Function of the Illustrative Scenario

The function of this scenario is methodological rather than evidentiary. It demonstrates that STR:

1. Can be implemented without advanced adaptive technology
2. Does not rely on learner self-regulation to initiate autonomy
3. Makes the moment of support withdrawal explicit and observable
4. Preserves experimental control by isolating a single design variable

By providing a concrete instantiation of the framework, the scenario addresses a common limitation of purely conceptual proposals: the absence of procedural clarity. At the same time, it avoids making claims about learning outcomes, effectiveness, or pedagogical superiority.

4.5.7 Summary

This illustrative scenario shows how the STR framework can be realized using a single text and a minimal sequence of design constraints. The example does not argue that supported reading followed by unsupported rereading is optimal, effective, or superior to alternative approaches. Instead, it demonstrates how categorical support withdrawal can be operationalized as a researchable event within L2 reading.

The value of the scenario lies in making the transition from supported to autonomous processing visible, controllable, and measurable—thereby enabling future empirical investigation of a design feature that is commonly assumed but rarely specified in CALL research.

5. Illustrative Measurement Possibilities (Protocol-Bound)

5.1 Protocol-Bound Descriptors Rather Than Psychological Constructs

The present framework does not introduce new psychological constructs of autonomy, dependency, or proficiency. Nor does it seek to operationalize existing constructs in a psychometric sense. Instead, STR specifies a set of protocol-bound descriptors that serve a purely descriptive function within the confines of the design.

Concepts such as autonomy or L1 dependency, when treated as psychological constructs, typically refer to relatively stable dispositions, involving motivational, strategic, and self-regulatory dimensions that generalize across tasks and contexts. STR makes no claims at this level. Any autonomy-related or dependency-related descriptor derived from the protocol reflects only a learner's adjustment to a specific, enforced design constraint under tightly controlled conditions.

For this reason, labels such as Autonomous Strength (AS) and L1 Dependency (L1-D) are introduced solely as protocol-internal descriptors. They do not represent latent variables, abilities, or traits, and they are not intended to function as indices in the psychometric or inferential sense. Their role is limited to naming observable patterns that emerge at the transition between supported and unsupported phases within the STR sequence.

No attempt is made to compute, validate, or generalize these descriptors in the present work. They are neither calibrated against external benchmarks nor proposed as transferable measures. Their sole purpose is to support structured description of how learners respond, within the protocol, to the categorical withdrawal of support.

5.2 Observable Data Anchors

STR relies exclusively on observable and minimally inferential data anchors that can be aligned with the enforced transition between phases. These anchors are not specific to STR, nor are they theoretically diagnostic on their own. Their relevance arises only from their placement within a tightly constrained design sequence.

Such data anchors may include:

- **Comprehension outcomes**, assessed through task-appropriate checks following each phase
- **Reading time**, recorded at the level of the full task and, where relevant, separately for each phase
- **L1 access behavior**, in implementations that allow logging of attempted interactions with support resources
- **Subjective cognitive load**, collected via brief self-report instruments immediately after each phase

Importantly, these data anchors are not interpreted as indicators of learning, proficiency, or strategy use in general. They are used solely to describe phase-to-phase adjustments under conditions where text, content, and prior resolution are held constant. Outside the STR protocol, the same data points would not carry the same interpretive meaning.

5.3 Transition-Based Descriptors

For descriptive convenience, this framework introduces two shorthand labels: an L1 dependency descriptor (L1-DI) and an autonomy adjustment descriptor (AS). These labels are used solely to refer to phase-difference descriptions within STR-controlled tasks. They are not indices or psychological constructs, and they are not intended to imply underlying latent variables.

The present manuscript does not specify formulas, validation procedures, reliability estimates, or normative interpretations for these descriptors. Such specifications fall outside the scope of this design framework and would only be defined within an empirical implementation study.

Within the STR protocol, the L1 dependency descriptor refers to observable differences in reliance on L1-mediated support between the supported and unsupported phases. The autonomy adjustment descriptor refers to changes in task-level performance across the transition from supported to unsupported reading. Both descriptors are contingent on the STR design and lose interpretive relevance outside it.

5.4 Interpretive Boundaries

The STR framework explicitly limits what its indices can claim.

They do **not** support inferences about:

- stable learner traits
- overall language proficiency
- long-term learning outcomes

They are intended solely for **protocol-internal analysis**, enabling researchers to describe how learners adjust to enforced autonomy under tightly controlled conditions.

By maintaining these boundaries, STR avoids construct overreach and positions its indices as descriptive tools for design-based research rather than as substitutes for validated psychometric measures.

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6. Cognitive Rationale (Minimal)

This section outlines the cognitive rationale underlying the STR framework. The goal is not to advance new cognitive theory, but to demonstrate that the design choices are compatible with established accounts of processing and learning, while remaining deliberately conservative in their claims.

6.1 Load Redistribution Across Phases

The two-phase structure of STR reflects a deliberate redistribution of cognitive load across time rather than an attempt to optimize all processing demands simultaneously.

In **Phase I**, the availability of in-flow support reduces **extraneous cognitive load** associated with search, navigation, and fragmentation of attention. By minimizing the need to consult external resources or interrupt reading flow, Phase I allows learners to allocate working memory resources to constructing a coherent representation of the text. This phase is designed to stabilize comprehension rather than to induce effortful processing.

In **Phase II**, the categorical removal of support increases **germane cognitive load** by requiring learners to sustain comprehension without external assistance. Because the text itself remains unchanged, the additional effort is not attributable to increased intrinsic difficulty, but to the need to rely on internally available representations. The framework therefore treats effort in Phase II as a consequence of support withdrawal rather than as a function of new linguistic input.

Importantly, STR does not assume that higher germane load necessarily leads to better learning outcomes. Load redistribution is presented as a design property that makes different types of processing demands observable across phases.

6.2 Retrieval Demand (Careful Framing)

The unsupported rereading phase in STR imposes **retrieval-like processing demands**, insofar as learners must reconstruct meaning without access to previously available cues. This requirement moves processing beyond simple recognition, which dominates when support remains visible, and toward active reliance on memory traces established during Phase I.

However, STR deliberately frames this demand as a **lower bound of retrieval practice**. The framework does not claim to implement delayed retrieval, spaced practice, or conditions known to optimize long-term consolidation. Immediate rereading is likely to draw on a mixture of working memory and emerging long-term representations, and the design makes no assumptions about the durability of the resulting traces.

For this reason, retrieval is treated as a **plausible processing mechanism**, not as an outcome claim. The framework does not equate successful Phase II performance with durable learning, nor does it assert that STR replaces established retrieval-based learning paradigms. Its contribution lies in embedding retrieval-like demands within the reading process in a controlled and observable manner.

6.3 Why Immediate Rereading Is Used

The use of immediate rereading in STR is motivated by considerations of **experimental control**, not by claims of pedagogical optimality.

By placing Phase II directly after Phase I, the framework minimizes variability associated with forgetting, interference, and contextual change. This temporal proximity allows researchers to isolate the effects of **support withdrawal itself**, rather than conflating them with time-based decay or spacing effects.

At the same time, the framework explicitly acknowledges the limitations of immediate rereading. Because the delay is minimal, Phase II performance may reflect short-term accessibility rather than stable retention. STR therefore does not claim to model consolidation processes, but instead offers a controlled baseline from which delayed or spaced variants could be systematically explored in future research.

7. Design Implications for CALL Systems

Beyond its value as a research framework, STR highlights several implications for the design of CALL systems that aim to support the development of reading autonomy.

7.1 Why Learner-Controlled Support Is a Confound

Many CALL systems rely on **learner-controlled support**, allowing users to decide when and whether to access glosses or dictionaries. While this approach aligns with principles of personalization and user agency, it also introduces a persistent confound for research and design.

When support is optional and cost-free, learners may systematically avoid effortful processing by defaulting to recognition and lookup strategies. In such cases, observed comprehension reflects interface efficiency rather than autonomous language processing. Consequently, it becomes difficult to determine whether learners can read independently or are simply skilled at navigating support tools.

This tension highlights a broader distinction between **interface optimization and learning optimization**. Designs that minimize friction may improve usability without necessarily promoting the processing conditions associated with autonomy.

7.2 STR as a Constraint-Based Design Pattern

STR illustrates an alternative approach in which autonomy is **enforced through constraint** rather than encouraged through choice. By removing support categorically at a predefined point, the framework introduces **productive friction**—a temporary increase in effort that makes autonomous processing unavoidable.

From a design perspective, this constraint serves three functions. First, it prevents learners from circumventing unsupported processing. Second, it creates a clearly defined transition that can be observed and measured. Third, it aligns system behavior with instructional intent by ensuring that autonomy is not merely an aspirational outcome, but a required condition of engagement.

STR thus exemplifies a constraint-based design pattern that may be relevant beyond reading, particularly in CALL contexts where always-available assistance risks obscuring learner capability.

7.3 Possible Digital Implementations (Non-Required)

Although STR is not tied to a specific technological platform, it can be instantiated across a range of formats.

At a **static level**, the framework can be implemented in fixed materials that present supported and unsupported versions of the same text sequentially.

At a **semi-digital level**, systems may log reading time, support access, or subjective load without enforcing complex interactivity.

At a **fully digital level**, platforms could enforce phase transitions programmatically and capture fine-grained behavioral data.

These possibilities are noted to illustrate design flexibility rather than to prescribe implementation details. The framework remains agnostic with respect to technology, focusing instead on the architectural role of enforced support withdrawal.

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8. Limitations and Boundary Conditions

The STR framework is intentionally constrained, and its scope is subject to several important limitations that delimit both interpretation and applicability.

First, the design relies on **massed practice**. The immediate transition from supported to unsupported reading means that Phase II occurs while representations from Phase I are still relatively accessible. As a result, observed performance in Phase II may reflect short-term accessibility rather than durable learning. STR therefore does not model spacing effects or long-term consolidation processes.

Second, the framework induces **immediate retrieval only**. While Phase II imposes retrieval-like demands, it does not instantiate delayed or repeated retrieval, which are known to play a central role in long-term retention. Any claims about consolidation or transfer would require additional phases or extended timelines beyond the current protocol.

Third, STR is primarily intended for **adult learners** engaged in self-directed or semi-guided reading. The binary nature of support withdrawal may not be appropriate for younger learners or contexts that require adaptive, instructor-mediated scaffolding.

Fourth, the framework targets **reading-focused outcomes only**. STR does not address productive skills such as speaking or writing, nor does it claim to generalize beyond reading comprehension and reading-related processing.

Finally, and most importantly, STR makes **no claims beyond the scope of the protocol itself**. The framework does not assert instructional superiority, nor does it claim to improve proficiency, retention, or autonomy in a general sense. Its contribution lies in design specification and measurement logic, not in outcome validation.

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9. Research Agenda

This agenda outlines design questions, not hypotheses. It does not specify predicted effects, expected advantages, or outcome hierarchies. Instead, it delineates directions in which the STR framework can be used to explore the feasibility, contrastability, and observability of enforced support withdrawal as a design variable.

At a general level, STR reframes the question of support withdrawal from one of instructional effectiveness to one of design contrast. The framework does not presuppose that enforced withdrawal differs positively or negatively from learner-controlled support environments. Its contribution lies in making such contrasts structurally explicit and analytically accessible, rather than in resolving them through outcome claims.

One design question concerns the temporal placement of unsupported rereading. The present framework specifies an immediate transition from supported to unsupported reading in order to maximize control and interpretability. Alternative instantiations may introduce delayed transitions between phases, not to test predictions about spacing effects, but to examine whether the observability of adjustment to support withdrawal is preserved under different temporal constraints.

A second design question involves the structure of support fading. STR adopts binary withdrawal to isolate the transition as a single, categorical design event. Other fading schemes—such as gradual or performance-contingent withdrawal—may be explored as design contrasts relative to this binary baseline. The purpose of such contrasts is not to establish superiority, but to assess how different fading architectures shape the visibility and interpretability of learner adjustment within the protocol.

The framework also supports contrasts between enforced withdrawal and learner-controlled support availability. By holding text and task constant while varying only the controllability of support, STR allows examination of how architectural constraints influence observable behavior, effort allocation, and support reliance. The framework remains deliberately agnostic as to which configuration is preferable, treating both as alternative design instantiations rather than competing interventions.

Beyond single-episode implementations, STR can be instantiated in repeated or longitudinal sequences comprising multiple supported–unsupported cycles. Such extensions are relevant not for tracking learning gains, but for examining the stability, variability, or evolution of adjustment patterns across repeated exposure to the same design constraint.

Finally, fine-grained process-oriented tools—such as interaction logging, time-based indicators, or eye-tracking—may be incorporated in STR-based work as optional observational aids. These tools are not integral to the framework and are not required for its application. Their role, where used, is limited to enhancing descriptive resolution of protocol-internal behavior rather than validating theoretical constructs or outcome effects.

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10. Conclusion

This article has presented Supported Two-Phase Reading (STR) as a design framework for formalizing the withdrawal of support in L2 reading. Rather than proposing an instructional method or advancing claims about learning effectiveness, STR specifies a minimal, protocol-level design that renders the transition from supported to unsupported processing explicit and analytically accessible.

STR reframes the core challenge in assisted reading research. The issue is not how to cultivate learner autonomy, but how to architecturally instantiate the condition of autonomy in order to observe it. By imposing support withdrawal as a mandatory design constraint, the framework shifts analytic focus away from whether support is available and toward how learners adjust when autonomous processing is the only available mode.

The central contribution of STR is therefore methodological. The framework offers a design grammar for the withdrawal event—a grammar that allows researchers to isolate and examine the transition from supported to unsupported processing as a distinct, manipulable design variable. Although such transitions have been ubiquitous in theoretical discussions of autonomy, they have remained largely absent from methodological specification in CALL research.

Crucially, the present work intentionally stops short of empirical instantiation. This is not a limitation to be remedied within the manuscript, but a defining boundary condition of its contribution. The value of

STR lies in design formalization rather than outcome evaluation, and its relevance should be assessed in terms of the analytic access it provides to an otherwise under-specified transition in L2 reading research.

Any empirical implementation of STR necessarily constitutes a subsequent step, external to the present framework, and subject to independent methodological choices and evaluative criteria. By clarifying the design conditions under which enforced support withdrawal becomes observable, STR aims to support such future investigations without presupposing their outcomes.

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